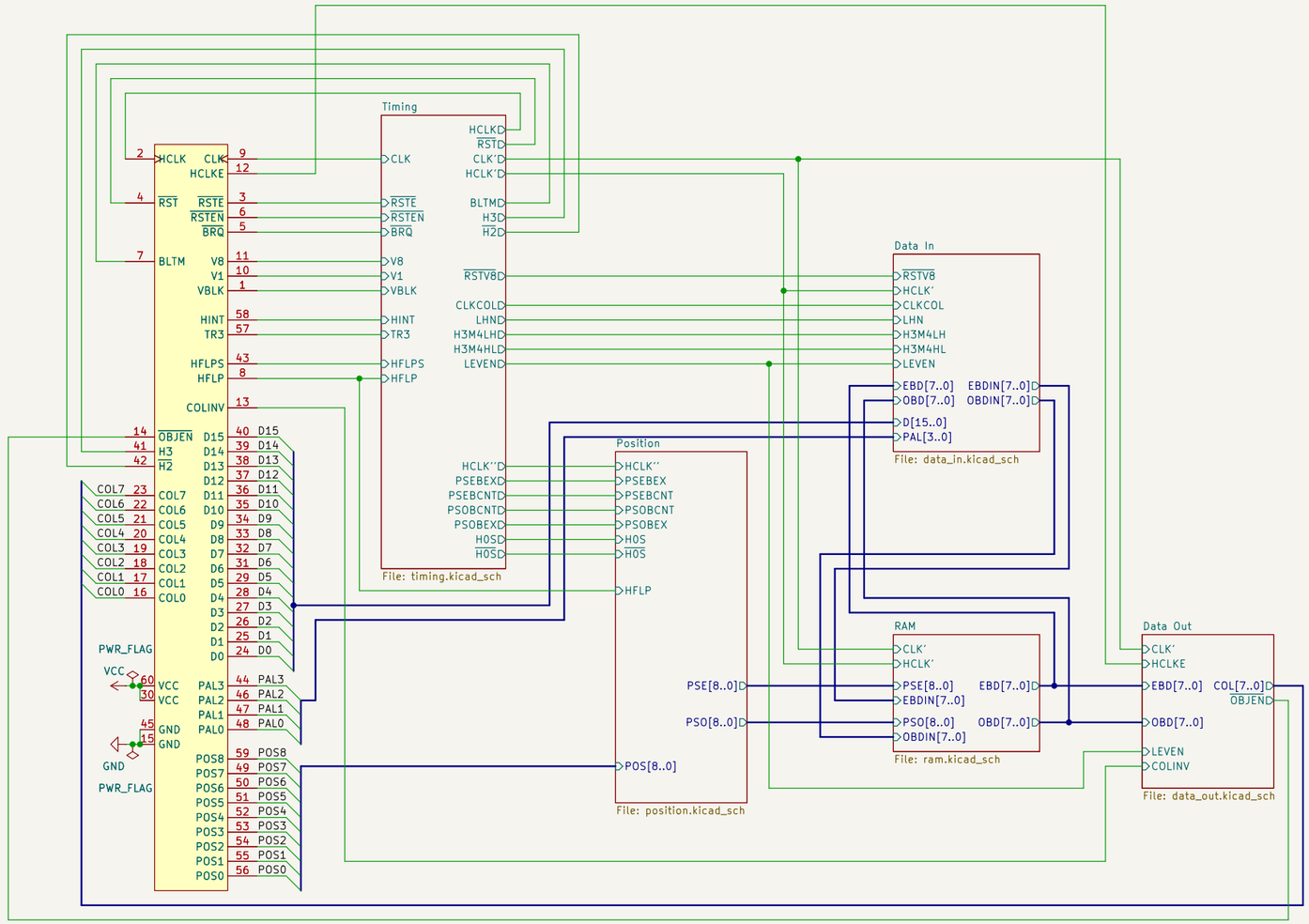


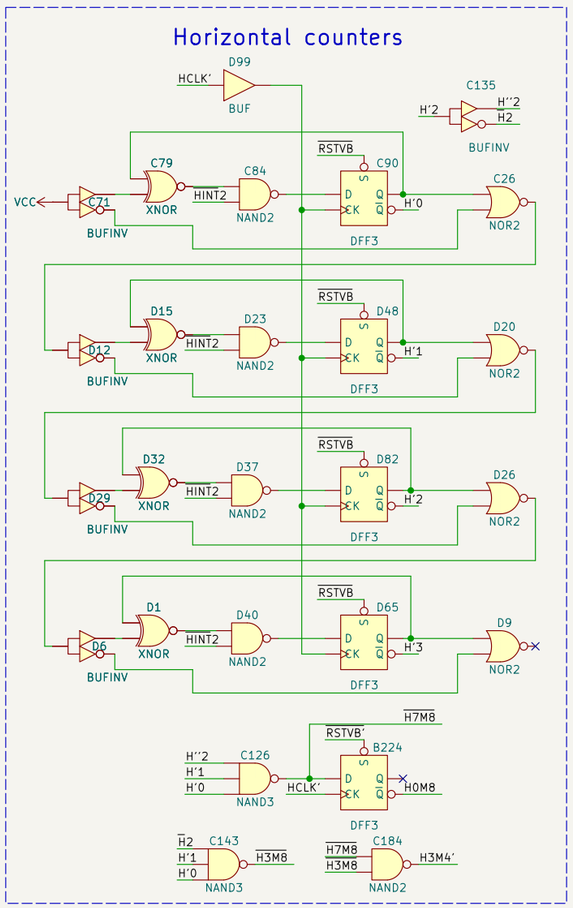
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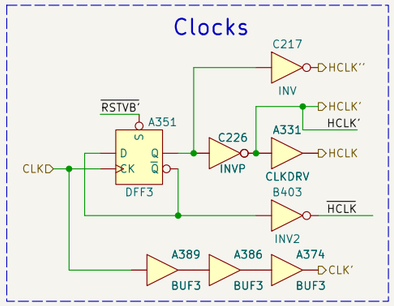
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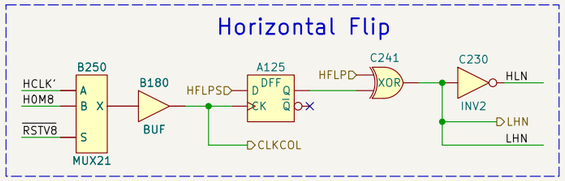
Horizontal counters



Clocks



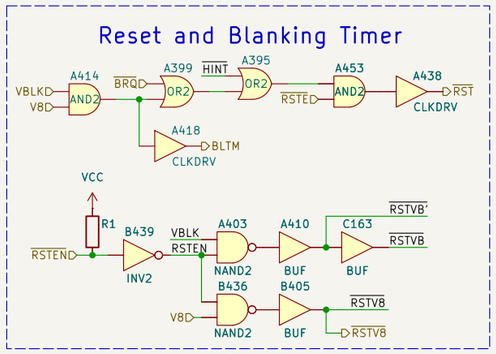
Horizontal Flip

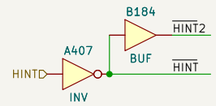


LHN = Low to High Nibble data in
HLN = High to Low Nibble data in

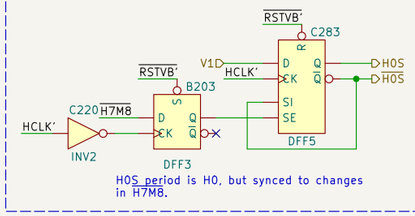
LHN is active when HFLP is not equal to HFLPS.

Reset and Blanking Timer



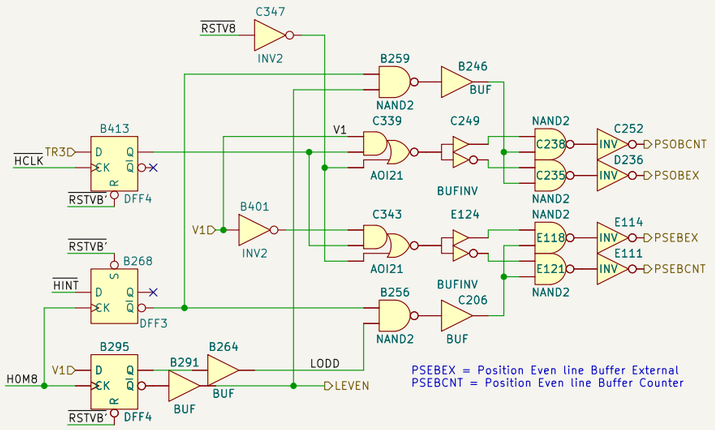


H0 Synced



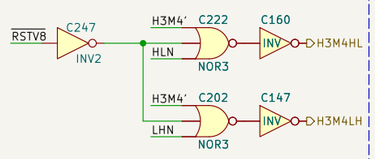
H0S period is H0, but synced to changes in H7M8.

Line Buffer timing



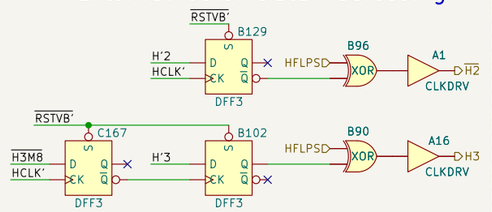
PSEBEX = Position Even line Buffer External
PSEBCNT = Position Even line Buffer Counter

Data In Timing



H3M4 is active at horizontal counter 3 and every 8 count after that. It is modulated by 8. H3M4HL is the same when horizontal flip is active.

External Color Data Addressing



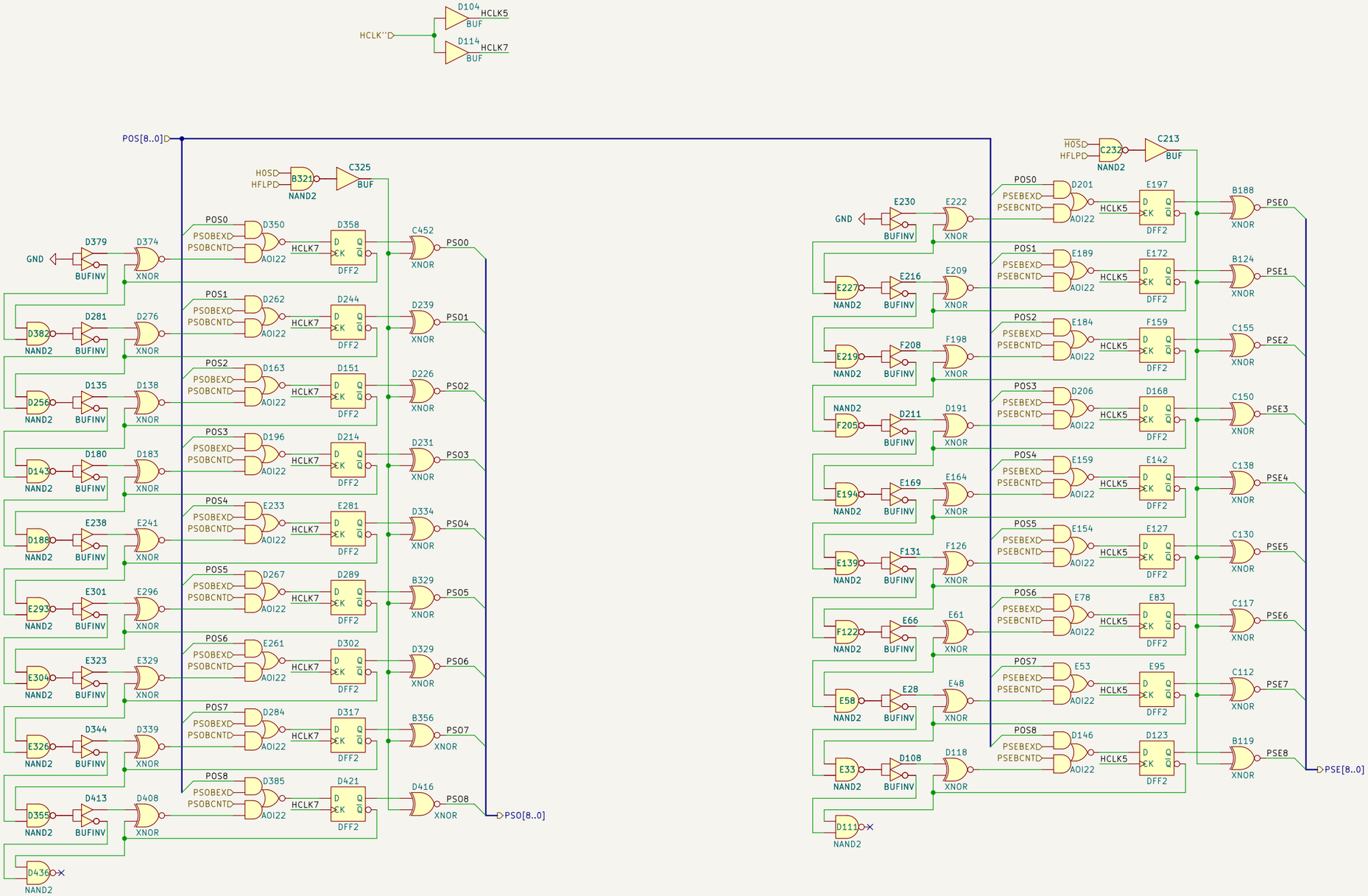
These counters make the external addressing in order 2,1,0,3

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If HLN is active usually C input is active, except every fourth pixel count when D becomes active.

HLN active

H3M4HL	H3M4LH	X
1	0	C
1	1	D

LHN active

H3M4HL	H3M4LH	X
0	1	B
1	1	D

Line Buffer Even Set

When
In order: D15, D11, D7, D3 -> DI[3:0]
Out order DO[3:0] -> [D3,D7,D11,D15]

When buffer has been set to ones, new data is enabled.

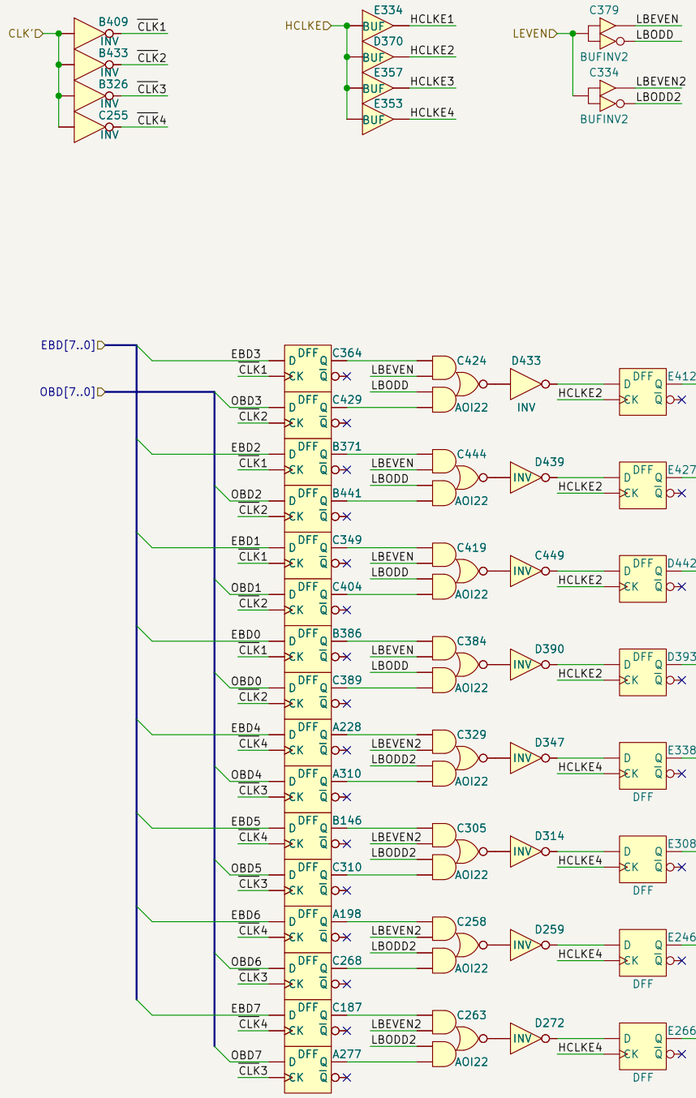
OBDNW = Odd Buffer Data New

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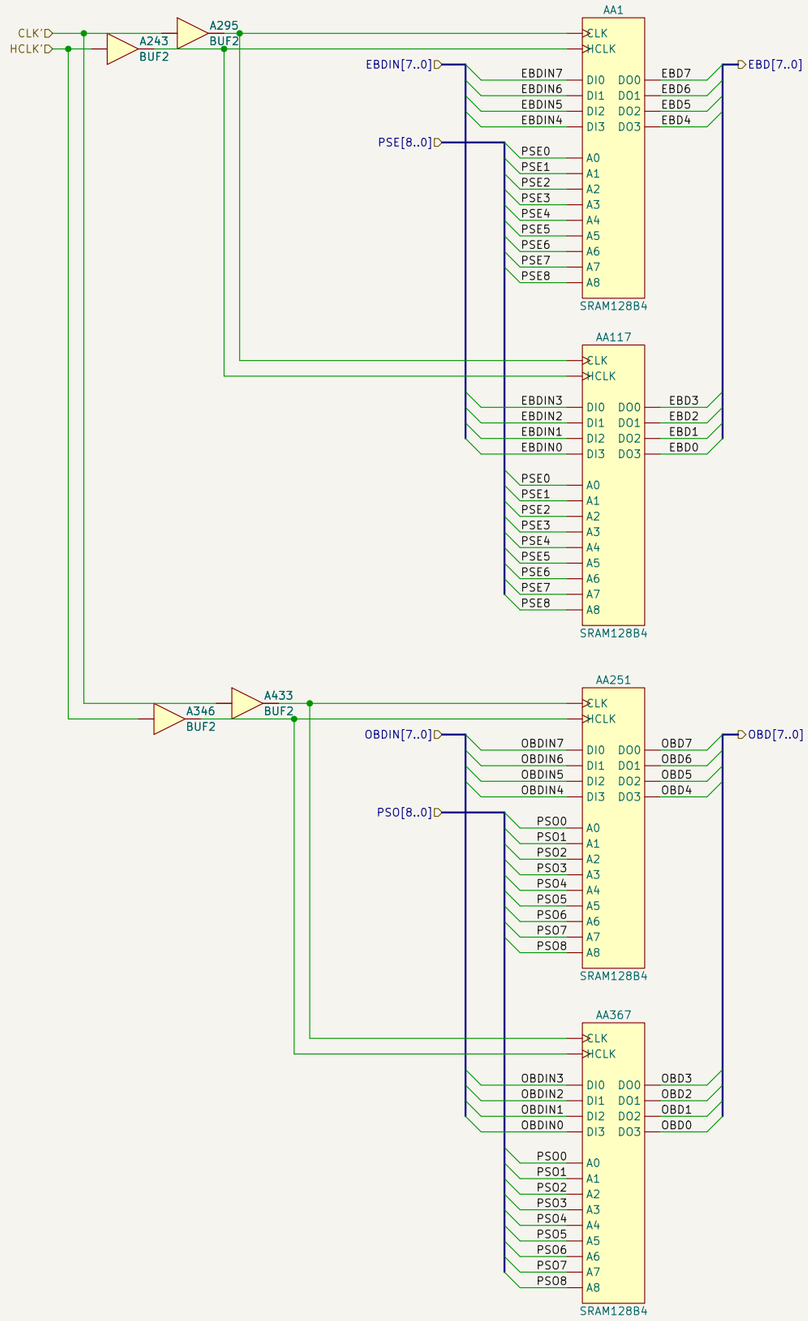
Sheet: /Data In/
File: data_in.kicad_sch

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Sheet: /Data Out/ File: data_out.kicad_sch		
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Sheet: /RAM/
File: ram.kicad_sch

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